
InstEmb: Instruction-Following Embeddings through Glimpses of the Future

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Abstract

Recent advances have empowered large language models (LLMs) with remarkable fine-grained instruction-following capabilities in text generation tasks. However, embedding methods typically rely solely on the hidden state of the input’s last token, limiting their ability to capture complete semantic signals distributed across the full output tokens. Moreover, existing discrete-to-continuous re-encoding approaches introduce semantic discontinuity. To address these limitations, we propose **InstEmb**, a novel instruction following embedding framework. InstEmb jointly optimizes two key aspects: (1) Input-Intrinsic semantic information, achieved by employing contrastive learning focused on the representation of the last input token, and (2) Output-Aware semantic information, captured through representation self-distillation leveraging learnable look-ahead tokens without introducing additional decoding latency. Additionally, we introduce **Dual-Anchor Alignment Pooling (DAAP)**, explicitly aligned with our dual training objectives. Extensive experiments demonstrate that InstEmb achieves state-of-the-art performance across multiple instruction following benchmarks without benchmark-specific supervised data.

1. Introduction

Recent advances in large language models (LLMs) have significantly enhanced their capability to follow fine-grained instructions, enabling remarkable zero-shot performance across diverse downstream tasks (Wang et al., 2023; Shen et al., 2023; Ji et al., 2023; Zhang et al., 2026b; Naveed et al., 2025). While instruction-following abilities have been extensively leveraged in text generation tasks, it remains challenging to achieve similar fine-grained instruction adapt-

ability when utilizing LLMs to produce text embeddings. Ideally, embeddings generated under different instructions should capture distinct semantic aspects of the same input text, providing richer and more targeted semantic representations (Li et al., 2025; Weller et al., 2025; Li et al., 2024a; Weller et al., 2024).

For instance, in product retrieval, embedding methods face challenges with queries exhibiting minor semantic variations. Consider two similar queries for a tent: “*Is this tent durable for outdoor use?*” and “*Is this tent compact for outdoor use?*”. While both concern outdoor suitability, their emphasis differs—durability versus portability. Embeddings should reflect these nuances within the product detail. Critically, while traditional encoders struggle when queries share high semantic overlap, due to their inability to capture fine-grained instruction shifts. This limits their effectiveness in instruction following retrieval.

Prior research indicated that aligning the embedding space with key tokens enhances semantic interpretability and instruction-following performance. Specifically, (Nie et al., 2025; Yamada & Zhang, 2025) demonstrated that LLM-derived embeddings inherently exhibit strong alignment with key tokens, while (Peng et al., 2024) showed that fine-tuning towards key tokens further improves semantic quality. Although previous works have shown the last token pooling approach achieves strong performance (Tang & Yang, 2024; Peng et al., 2024), it fundamentally conflates distinct semantic roles. The last token primarily captures the semantics of the input and instruction but fails to incorporate the semantic information of output, as the latter is distributed across multiple critical tokens and cannot be adequately aggregated into a single position. To clarify this distinction, we hereafter refer to these two roles as **Input-Intrinsic Semantics** and **Output-Aware Semantics**.

In the realm of Retrieval-Augmented Generation (RAG), recent attempts to enhance Output-Aware Semantics capabilities through re-encoding of output tokens or hypothetical documents (Gao et al., 2023; Ma et al., 2023) introduce another limitation: The method enacts discrete-to-continuous conversion through autoregressive token generation followed by re-encoding. This dual-stage process creates a semantic reconstruction gap, where sampling during decoding severs continuity in latent representations.

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These observations suggest that optimizing embedding results requires consideration of the following key aspects: 1) Both Input-Intrinsic Semantics and Output-Aware Semantics need to be used simultaneously, and the acquisition of Output-Aware Semantics should not incur additional time consumption due to decoding. 2) Both types of semantics should ideally be aggregated and preserved within the LLM’s latent space, rather than relying on re-encoding or discrete-to-continuous mapping.

To address these limitations, we propose a novel instruction following embedding framework, **InstEmb**. Our method explicitly optimizes both types of information: Input-Intrinsic Semantics captured by the last input token, and Output-Aware Semantics distributed across the suffix of learnable tokens, which we hereafter refer to as **look-ahead** tokens. We adopt this terminology because the function of our learnable tokens is similar to the concept of speculative decoding field (Monea et al., 2023; Xiao et al., 2024b; An et al., 2025). The look-ahead tokens, which can be conceptually viewed as a form of suffix soft prompt, provide the model with implicit semantic previews of future tokens.

During inference, **InstEmb** generates embeddings solely through the prefilling stage, efficiently encoding output-related information without multi-step decoding. Furthermore, we propose **DAAP**, a pooling strategy explicitly aligned with our training objectives, eliminating empirical pooling selection and ensuring consistently optimal embedding performance.

We empirically validate InstEmb across multiple instruction following dense retrieval benchmarks, demonstrating strong effectiveness and efficiency compared to state-of-the-art baselines such as Inbedder (Peng et al., 2024) and FollowIR (Weller et al., 2025).

In summary, our contributions are:

- We propose **InstEmb**, a novel instruction following embedding framework utilizing learnable look-ahead tokens and embedding distillation to achieve fine-grained, sentence-level instruction adaptation.
- We introduce a novel embedding distillation objective based on look-ahead tokens that effectively transfers instruction-following capabilities from a frozen instruction-tuned LLM.
- InstEmb attains state-of-the-art performance across multiple instruction following benchmarks without benchmark-specific supervised training.

2. Related Work

Instruction-tuned Embedding based on LLMs. Recent embedding approaches leverage large language mod-

els (LLMs) by fine-tuning them with instructions to enhance semantic representation quality and adaptability. Recent works predominantly utilize fixed, task-level instructions: LLM2Vec (BehnamGhader et al., 2024) transforms autoregressive decoders into bidirectional encoders for stronger representational power; GRITLM (Muennighoff et al., 2024) unifies generative and representational tasks through instruction-tuning; E5-Mistral (Wang et al., 2024) employs contrastive tuning on synthetic query-document pairs; NV-Embed (Lee et al., 2025) optimizes embeddings via a two-stage contrastive learning paradigm; and ECHO (Springer et al., 2025) repeats input contexts to mitigate autoregressive limitations. Instructor (Su et al., 2023), while adopting contrastive training across numerous NLP tasks, still uses fixed task-level instructions rather than dynamically adapting instructions per instance.

Moving beyond task-level instructions, recent approaches dynamically adjust embeddings according to instance-level (per-query) instructions. BGE-icl (Li et al., 2025) and RARE (Tejaswi et al., 2024) leverage in-context learning to adapt embeddings using task-specific few-shot examples; PIE (Li et al., 2024a) constructs contrastive examples guided by additional linguistic parsing; FollowIR (Weller et al., 2025) employs professional assessor narratives to interpret complex search intents; Promptriever (Weller et al., 2024) utilizes a large-scale, instance-level instruction dataset from MS MARCO; and INBEDDER (Peng et al., 2024) introduces an embed-via-answering paradigm, fine-tuning models on abstractive QA tasks conditioned on dynamic user instructions, significantly enhancing instruction-following capabilities.

Another category adapts LLM embeddings under specific conditions. For instance, GSTransformer (Feng et al., 2025) realigns pre-computed embeddings with user instructions. Hyper-CL (Yoo et al., 2024) transforms condition embeddings into projection layers to adapt sentence embeddings. CASE (Zhang et al., 2026a) generates condition-aware embeddings through attention-based conditioning and supervised projection.

Look-Ahead Token Usage Despite the differing terminology, the concept of look-ahead tokens has found applications in various fields. For speculative decoding, Monea et al. (2023) used special tokens to enable parallel decoding without auxiliary models; Xiao et al. (2024b) trained parallel drafters with placeholders to predict multiple future tokens; and An et al. (2025) proposed a target-agnostic drafting framework with token discarding. For implicit reasoning, Wu et al. (2025) used implicit thought tokens with Jacobi iteration to reduce latency. In non-autoregressive generation, Nie et al. (2026) employed mask tokens and diffusion denoising for fully parallel generation.

Knowledge Distillation Knowledge distillation has been widely employed to transfer semantic knowledge from powerful teacher models to smaller student embedding models. Early methods, such as DistilBERT (Sanh et al., 2019), TinyBERT (Jiao et al., 2020), and MiniLM (Wang et al., 2020), primarily leverage logits or attention-based alignment. In addition, some contrastive distillation methods focus on optimizing representations, such as CRD (Tian et al., 2019), CKD (Xu et al., 2022), and SEED (Fang et al., 2021). These methods leverage distillation loss to capture semantic relationships among data, thereby enhancing the representation learning capability of the student model.

3. Methodology

As discussed in the Introduction, existing embedding methods exhibit two fundamental limitations: (1) relying solely on the input last token’s representation; (2) discrete-to-continuous re-encoding introduces a semantic reconstruction gap.

To overcome these limitations, **InstEmb** employs learnable look-ahead tokens appended directly to the input text, enabling continuous representation self-distillation from a frozen teacher model in order to get Output-Aware semantic. It also leverages standard contrastive learning at the input last token position to maintain the performance.

3.1. Architecture

Given an autoregressive LLM with input sequence $\mathbf{x}$ (input text concatenated with instruction) and target output $\mathbf{y}$, we define learnable look-ahead tokens $\mathbf{s} = [s_1, s_2, \dots, s_L]$ to form the training inputs:

$$\mathbf{x}_{\text{student}} = [\mathbf{x}; \mathbf{s}], \quad \mathbf{x}_{\text{teacher}} = [\mathbf{x}; \mathbf{y}_{\text{trunc}}]$$

where $\mathbf{Y}_{\text{trunc}}$ represents the first (L) tokens of $\mathbf{Y}$. Crucially, unlike conventional knowledge distillation that relies on a superior teacher model, InstEmb employs **isomorphic self-distillation**: both the student and teacher share the identical architecture and initialization, with the teacher frozen to provide *output-conditioned supervision* during training. This architecture enables single-step inference without multi-step decoding overhead or semantic reconstruction gaps. The overall architecture of InstEmb is illustrated in Figure 1, and we elaborate on each component in the following parts.

3.2. Training Objectives

To ensure a coherent flow of understanding, we begin by describing our approach to optimizing the Output-Aware Semantics.

3.2.1. REPRESENTATION SELF-DISTILLATION FOR OUTPUT-AWARE SEMANTICS OPTIMIZATION

Inspired by knowledge distillation and speculative decoding (Li et al., 2024b; 2026), we employ position-wise distillation loss to align student look-ahead token representations with teacher target embeddings, enabling efficient transfer of Output-Aware semantic information that is captured by the whole output sequence.

Formally, let $\mathbf{h}_i^S, \mathbf{h}_i^T \in \mathbb{R}^d$ denote the student and teacher hidden states at position i ($i = 1, \dots, |\mathbf{x}| + L$). We consider two distillation strategies: directly aligning hidden states (MSE) and aligning probability distributions (KL), to explore semantic alignment at different granularities.

- **Mean Squared Error (MSE) Loss:** Direct alignment of hidden states:

$$\mathcal{L}_{\text{Distill}} = \frac{1}{L} \sum_{i=|\mathbf{x}|+1}^{|\mathbf{x}|+L} \left\| \mathbf{h}_i^S - \mathbf{h}_i^T \right\|_2^2 \quad (1)$$

- **Kullback–Leibler (KL) Divergence Loss:** Alignment of output probability distributions obtained via language modeling heads:

$$\mathcal{L}_{\text{Distill}} = \frac{1}{L} \sum_{i=|\mathbf{x}|+1}^{|\mathbf{x}|+L} \sum_{v \in \mathcal{V}} P_i^S(v) \log \frac{P_i^S(v)}{P_i^T(v)} \quad (2)$$

where P_i^S and P_i^T denote the student and teacher distributions, respectively. $\mathcal{V}$ is the vocabulary set

From an information-theoretic perspective, the distillation loss maximizes the mutual information between student look-ahead representations and teacher target embeddings. This enables efficient knowledge transfer from the teacher’s output-conditioned representations to the student’s input-only accessible tokens. The position-wise alignment ensures that each look-ahead token captures specific aspects of the target sequence semantics, creating a distributed representation of Output-Aware information that supplements the Input-Intrinsic semantic signal at the input’s last token.

3.2.2. CONTRASTIVE LEARNING FOR INPUT-INTRINSIC SEMANTIC OPTIMIZATION

While representation self-distillation optimizes Output-Aware Semantics from look-ahead tokens, recent embedding optimization methods (BehnamGhader et al., 2024; Springer et al., 2025) commonly utilize InfoNCE loss (Oord et al., 2018) to optimize Input-Intrinsic Semantics.

To enable multi-view contrastive learning, we construct four representations for each instance i by feeding specific sequences to the student and teacher models. First, we pass

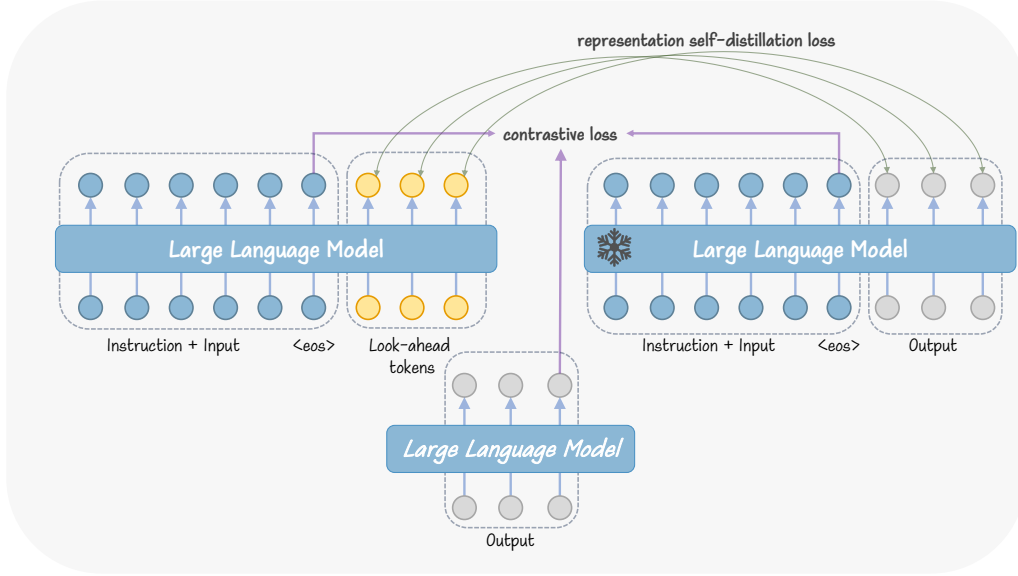


Figure 1. The overall architecture of InstEmb. It jointly optimizes two loss functions: a representation self-distillation loss to capture Output-Aware semantic signals, and a contrastive learning loss at the input’s last token position to enhance Input-Intrinsic semantic alignment explicitly. The definition of every view could be found in 3.2.2.

the input sequence x through the student model twice with distinct SimCSE-style dropout masks (Gao et al., 2021), extracting the hidden states at the last input position $|x|$ as $s_i^{(1)}$ and $s_i^{(2)}$ (i.e., $h_{|x|}^S$). Simultaneously, we feed x to the frozen teacher model and extract the hidden state at the same position as t_i (i.e., $h_{|x|}^T$). Additionally, we encode the target sequence y with the student model and extract the hidden state at the last output position $|y|$ as a_i (i.e., $h_{|y|}^S$). Formally, the view set for instance i is defined as $\mathcal{V}_i = \{s_i^{(1)}, s_i^{(2)}, t_i, a_i\}$.

We adopt an improved multi-positive contrastive loss based on the InfoNCE framework. Given a minibatch of N instances, the loss is:

$$\mathcal{L}_{CL} = -\frac{1}{4N} \sum_{i=1}^N \sum_{z \in \mathcal{V}_i} \log \frac{\sum_{z' \in \mathcal{V}_i \setminus \{z\}} \exp(\text{sim}(z, z')/\tau)}{Z_i(z)}, \quad (3)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, τ is the temperature, and $Z_i(z)$ is the normalization factor aggregating all negative pairs:

$$Z_i(z) = \underbrace{\sum_{z' \in \mathcal{V}_i \setminus \{z\}} \exp(\text{sim}(z, z')/\tau)}_{\text{intra-instance positives}} + \underbrace{\sum_{j \neq i} \sum_{z'' \in \mathcal{V}_j} \exp(\text{sim}(z, z'')/\tau)}_{\text{inter-instance negatives}}. \quad (4)$$

The numerator contrasts each anchor against all other views

from the same instance (positives), while the denominator additionally includes all views from other instances as negatives.

3.2.3. JOINT OPTIMIZATION

The final training objective jointly optimizes both Output-Aware and Input-Intrinsic semantic signals:

$$\mathcal{L}_{\text{InstEmb}} = \mathcal{L}_{\text{Distill}} + \mathcal{L}_{\text{CL}} \quad (5)$$

3.3. Dual-Anchor Alignment Pooling (DAAP)

InstEmb jointly optimizes Input-Intrinsic Semantics via contrastive learning at the last input token and Output-Aware Semantics via representation self-distillation on look-ahead tokens. However, standard pooling methods (e.g., last-token pooling or mean pooling) fail to explicitly integrate both optimized signals, capturing only a single semantic perspective.

To ensure theoretical alignment between training objectives and inference-time embedding extraction, we propose **Dual-Anchor Alignment Pooling (DAAP)**, which is explicitly designed to fuse two Output-Aware semantic anchors without the need for empirical pooling selection:

- **Input-Intrinsic Anchor:** The hidden state of the final input token.
- **Output-Aware Anchor:** The averaged hidden states of the look-ahead tokens.

Formally, DAAP computes the final embedding as follows:

$$e = \frac{1}{2} \left(h_{|x|}^S + \frac{1}{L} \sum_{j=|x|+1}^{|x|+L} h_j^S \right) \quad (6)$$

4. Experiments

In this section, we empirically validate the effectiveness of our proposed InstEmb framework. We first introduce our experimental setup, including datasets, baselines, and evaluation metrics. Then, we present our main results, demonstrating the strength of InstEmb over existing state-of-the-art methods. Finally, we conduct extensive ablation studies to analyze the impact of different design choices.

4.1. Experimental Setup

4.1.1. DATASETS

We utilize the training dataset proposed by (Peng et al., 2024), comprising approximately 200,000 abstractive question-answer pairs from 11 diverse QA datasets with stopwords removed.

For evaluation, we assess InstEmb’s performance on two categories of benchmarks: (1) **Instruction-following embedding benchmarks**, including Inst.STSb, IntentEmotion, NYTCluster (Peng et al., 2024), and FollowIR benchmark (Weller et al., 2025), which evaluate the model’s ability to follow instructions and capture semantic similarity under various contexts; and InfoSearch benchmark (Zhou et al., 2025), which evaluates instruction-following capacities beyond content relevance. (2) **Generic sentence embedding tasks** from MTEB benchmark¹ including AskUbuntuDupQues, TwentyNewsgroups, SciDocsRR, and StackOverflowDup, which cover a wide range of semantic tasks for comprehensive embedding quality evaluation.

4.1.2. EVALUATION METRICS

We employ Spearman correlation for Inst.STSb, the harmonic mean of success rates for IntentEmotion, mean average precision (mAP) for AskUbuntu, SciDocs, and StackOverflow, and V-measure for clustering tasks (NYTCluster and 20NewsGroups). **FollowIR** benchmark employs nDCG@5 for News21, MAP@1000 for Core17/Robust04, and p-MRR (Weller et al., 2025) across all collections. normalized discounted cumulative gain at 5 (nDCG@5) measures ranking quality by cumulating the gains of retrieved documents, discounted logarithmically by their rank, and normalized by the ideal DCG. p-MRR metric ranges from -100 (indicating complete instruction contradiction) to 100

(reflecting perfect compliance). **InfoSearch** employs three metrics to evaluate performance: the Strict Instruction Compliance Ratio (SICR), the Weighted Instruction Sensitivity Evaluation (WISE) (Zhou et al., 2025), and p-MRR.

4.1.3. BASELINES

We compare against three categories of models: (1) **General LLMs**: Llama2 (Touvron et al., 2023), Llama3 (Grattafiori et al., 2024), Mistral-7b-instruct (Jiang et al., 2023), DIF-FEMBED (Zhang et al., 2025), and PonTE (Yamada & Zhang, 2025) (2) **Fixed-instruction embedding models**: LLM2Vec (BehnamGhader et al., 2024), GritLM (Muenighoff et al., 2024), ECHO (Springer et al., 2025), and e5 (Wang et al., 2022); (3) **Instruction-adaptive models**: Instructor (Su et al., 2023), Inbedder (Peng et al., 2024), FollowIR (Weller et al., 2025), Promptriever (Weller et al., 2024).

Further implementation details and hyperparameters could be found in Appendix A.1.

4.2. Main Results Analysis

Table 1 and Table 2 present the Input-Intrinsic evaluation results comparing our proposed InstEmb method against several baseline embedding methods across two categories of benchmarks: instruction-following embedding tasks and generic sentence embedding tasks.

Instruction-Following Embedding Tasks. Our InstEmb achieves new state-of-the-art performance across instruction-following benchmarks, demonstrating significant improvements in instruction comprehension. On the FollowIR benchmark (Table 1), our InstEmb achieves 28.5 average score and +15.6 p-MRR ratio - significantly surpassing previous SOTA methods like FollowIR-7B (24.8 score/+12.2 p-MRR) and Promptriever (26.1 score/+11.2 p-MRR). Remarkably, while these competitors use supervised training data specifically optimized for FollowIR, our model achieves strong performance through zero-shot inference without any benchmark-specific training. This result highlights InstEmb’s strong generalization capability in instruction following retrieval.

On the InfoSearch benchmark, InstEmb_{MSE} also demonstrates competitive performance, achieving a mean p-MRR of 7.7 with 20.1 SICR and 13.3 WISE scores. This represents a substantial improvement over the base Llama-3-8B-instruct model (6.6 p-MRR) and outperforms other strong baselines including FollowIR-7B (4.1 p-MRR). The consistent superiority across both FollowIR and InfoSearch benchmarks further validates InstEmb’s robust instruction-following capabilities in diverse retrieval scenarios.

Beyond above, InstEmb_{MSE-DAAP} attains a mean score of

¹The MTEB leaderboard can be found at <https://huggingface.co/spaces/mteb/leaderboard>

InstEmb

Model	FollowIR								InfoSearch		
	Robust04		News21		Core17		Mean		Mean		
	MAP	p-MRR	nDCG	p-MRR	MAP	p-MRR	Score	p-MRR	SICR	WISE	p-MRR
GritLM-Reranker	9.7	+6.1	10.2	+3.4	9.8	+8.6	9.9	+6.0	6.9	-11.1	-4.3
Mistral-7B-instruct	23.2	+12.6	27.2	+4.8	19.7	+13.0	23.4	+10.1	0.0	-49.2	-32.4
DiffEMBED	18.9	+5.7	27.7	+3.6	16.2	+6.0	20.9	+5.1	-	-	-
FollowIR-7B	24.8	+13.7	29.6	+6.3	20.0	+16.5	24.8	+12.2	12.5	13.4	4.1
Promptriever	28.3	+11.7	28.5	+6.4	21.6	+15.4	26.1	+11.2	-	-	-
Llama-2-7B-chat	6.3	+2.0	1.7	+0.2	5.4	+2.8	4.5	+1.7	8.4	-18.7	-10.9
InstEmb-MSE	10.8	+8.9	15.2	+0.0	9.5	+10.1	11.8	+6.3	10.5	-14.4	-8.4
Llama-3-8B-instruct	12.7	+2.3	17.6	-1.0	11.7	+1.7	14.0	+1.0	19.6	13.1	6.6
InstEmb-MSE	29.2	+19.1	32.3	+5.4	24.0	+22.4	28.5	+15.6	20.1	13.3	7.7

Table 1. Main results on FollowIR (left) and InfoSearch (right). InstEmb achieves state-of-the-art performance on FollowIR without extra supervised data tuned for this benchmark.

Model	I.STSb	IntEmo.	NYT.	Mean	AskU.	20news	SciD.	StackO.	Mean
LLM2Vec-llama2-7b	-	-	-	-	63.13	51.04	84.03	51.02	62.30
Echo-mistral-7b-eos	-	-	-	-	64.1	53.04	83.68	51.84	63.16
instructor-large	-15.02	47.96	49.96	27.63	63.48	53.51	81.83	50.50	62.33
e5-large-v2	0.00	30.24	50.07	26.77	59.01	47.94	83.84	50.60	60.35
llama2-7b-chat _{InputLast}	23.24	95.56	56.87	58.55	55.70	47.43	76.58	41.78	55.37
Inbedder	22.07	89.68	64.65	58.80	60.32	52.33	80.61	44.77	59.51
InstEmb-MSE _{DAAP}	23.51	93.45	70.77	62.57	60.16	52.56	80.43	47.37	60.13
llama3-8b-instruct _{InputLast}	44.2	92.8	25.6	54.2	58.3	54.0	79.3	44.6	59.0
PonTE	44.60	89.50	48.03	60.71	57.49	52.71	82.47	41.97	58.66
Inbedder	23.1	94.5	62.0	59.9	61.9	54.8	83.2	46.2	61.52
InstEmb-MSE _{DAAP}	41.37	94.12	65.76	67.08	63.25	54.26	84.86	48.40	62.69
InstEmb-KL _{DAAP}	39.24	94.88	62.32	65.48	63.34	55.39	85.50	49.35	63.39
Qwen2.5-7B-Instruct _{InputLast}	44.34	94.87	59.24	66.14	53.72	52.72	77.99	37.43	55.46
InstEmb-MSE _{DAAP}	41.60	96.63	64.88	67.70	57.60	53.56	82.16	41.62	58.73
InstEmb-MSE _{DAAP} (14B Teacher)	42.05	96.57	67.64	68.75	58.09	53.23	82.69	41.74	58.94

Table 2. Main results on Inst.STSb, IntentEmotion, NYTCluster, and all generic sentence embedding tasks. Subscripts “InputLast” and “DAAP” indicate the pooling method; Pooling methods’ details can be found in the Appendix A.2. Inbedder under llama3-8b-instruct indicates our re-implementation of Inbedder(Peng et al., 2024).

67.08% on Inst.STSb, IntentEmotion, and NYTCluster. It gets an absolute improvement of 7.18% over the SFT baseline (59.9%). Consistent gains are also observed on llama2, where InstEmb_{MSE-DAAP} improves the mean score by 3.77% over the SFT-based Inbedder (62.57% vs. 58.80%). In contrast to fixed-instruction embedding models (e5, instructor) that fail to model dynamic instructions effectively, InstEmb demonstrates robust advantages across all instruction-following embedding tasks.

General Sentence Embedding Tasks. Despite being optimized on instruction-following data, InstEmb demonstrates strong performance on general embedding benchmarks, achieving a competitive mean score of 63.39% (InstEmb-KL_{DAAP}) with 7× less training data than specialized baselines like LLM2Vec and Echo-mistral (both 1.5M examples). The performance of InstEmb on general sentence embedding tasks highlights the transferability and generalization ability of our instruction-based embedding method, an advantage also observed in its strong zero-shot performance on the FollowIR benchmark.

Backbone and Teacher Generalization. We further evaluate InstEmb on Qwen2.5-7B-Instruct to test whether its effectiveness transfers beyond the Llama family. InstEmb-MSE_{DAAP} improves both instruction-following and general embedding performance, increasing the two mean scores from 66.14%/55.46% to 67.70%/58.73%. With a larger teacher(Qwen2.5-14B-Instruct), the instruction-following mean further increases to 68.75%, while the general-task mean remains slightly higher at 58.94%. Since the larger teacher and student differ in hidden-state dimensionality and representation scale, we use a lightweight one-layer MLP to align the teacher representations with the student hidden-state space.

Overall, the results show that MSE-based self-distillation is particularly effective for instruction-following embedding tasks, while KL-divergence performs better on generic sentence embedding tasks. The Qwen2.5 experiments further indicate that InstEmb is broadly applicable across backbone architectures and can benefit from stronger teachers with proper hidden-state alignment. The following ablation study provides a detailed analysis of these effects.

5. Ablation Studies & Analysis

5.1. Effectiveness of Representation Self-Distillation

To validate the effectiveness of our representation self-distillation approach and understand the contribution of different loss functions, we conduct a controlled ablation study. Specifically, we remove $\mathcal{L}_{CL}$ and compare the performance of different training objectives applied to look-ahead tokens, allowing us to isolate the impact of distillation strategies on Output-Aware semantic learning.

Experimental Setup. We evaluate four training configurations: **SFT** (standard autoregressive supervised fine-tuning without look-ahead tokens), **CE** (cross-entropy loss applied to look-ahead tokens without teacher guidance), **MSE** (eq 1), and **KL** (eq 2). For distillation methods (MSE and KL), the teacher model receives the full input sequence and gold output, while the student model receives only the input and learnable look-ahead tokens. The distillation loss replaces the standard cross-entropy on look-ahead tokens, enabling the student to capture output-aware semantics through teacher guidance rather than direct token-level supervision. As shown in Table 3, several key observations emerge:

Distillation Superiority: Both MSE and KL distillation methods significantly outperform the CE baseline, demonstrating the effectiveness of teacher-guided semantic transfer. Confirming that self-distillation enables more effective Output-Aware semantic than imitation target.

Task-Specific Specialization: The two distillation objectives exhibit Output-Aware semantic strengths. KL divergence excels in generalizable tasks (higher Mean Gen.), likely due to its ability to preserve instruction-agnostic knowledge through probability distribution alignment. Conversely, MSE demonstrates superior performance on instruction-specific tasks (higher Mean Inst.), suggesting its effectiveness in fine-grained task nuance modeling through direct representation regression.

	Inst.	Gen.	Score	p-MRR
SFT	59.9	61.52	5.0	+8.4
CE	60.13	61.93	27.3	+14.8
KL	63.41	63.01	27.9	+14.9
MSE	64.16	61.81	26.7	+15.0

Table 3. Ablation study comparing different training objectives without contrastive learning.

5.2. impact of look-ahead token length

we ablate the length of look-ahead tokens. During the inference phase, we employ different look-ahead lengths, specifically set to 0, 1, 4, and 8. When the look-ahead length is

set to 0, it is equivalent to using only the last input token.

As shown in Fig. 2, it can be observed that almost all tasks exhibit a significant performance jump as the look-ahead token length increases from 0 to 1. Except several tasks such as IntSTS and intEmo, because these tasks were designed with very short output, resulting in high information density, the improvement brought by Output-Aware Semantics is not significant and may even decrease.

Based on these observations, we conclude that the presence of look-ahead is crucial, but the performance is not highly sensitive to the specific number of tokens added. However, we recommend longer sequences for tasks like NYTCluster that benefit from extended look-ahead.

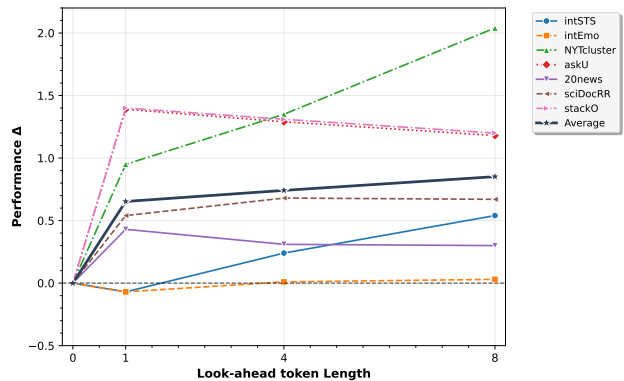


Figure 2. ablation study on Look-ahead token length

5.3. Multi-View Contrastive Learning Analysis

Table 4 analyzes the contributions of different embedding views in the multi-view contrastive learning framework. Comparing the impact of removing the self-supervised dropout view ($s_i^{(2)}$; Δ -5.78) versus removing both teacher input and student output views (t_i and a_i ; Δ -1.75) highlights the critical role of the dropout-augmented view for contrastive stability, as its absence causes severe performance degradation and embedding collapse.

Retaining only the first student input view and the student output view ($-s_i^{(2)}$, $-t_i$; Δ -2.96) demonstrates that a simple dual-tower contrastive setup between input and output embeddings also serves as a viable strategy.

Furthermore, the teacher input view (t_i), closely aligned with the distillation objective, provides essential Output-Aware supervision; its removal (individually or jointly) consistently results in performance drops, underscoring its importance for overall embedding quality.

Strategy	Inst.	Gen.
Full view set ($\mathcal{V}_i$)	67.08	62.69
$-s_i^{(2)}$	61.30 (−5.78)	62.51 (−0.18)
$-t_i$	65.05 (−1.68)	61.01 (−2.03)
$-a_i$	65.54 (−1.54)	61.96 (−0.73)
$-s_i^{(2)}, -t_i$	64.12 (−2.96)	61.92 (−0.77)
$-s_i^{(2)}, -a_i$	56.44 (−10.64)	61.23 (−1.46)
$-t_i, -a_i$	65.33 (−1.75)	61.91 (−0.78)
−All views	64.16 (−2.92)	61.81 (−0.88)

Table 4. Ablation study on contrastive learning strategies. We remove views individually or jointly to measure their contributions. $-s_i^{(2)}$ denotes removing the second student input view (SimCSE-style augmentation), $-t_i$ denotes removing the teacher input view, and $-a_i$ denotes removing the student output view.

5.4. Impact of Pooling Methods

We evaluate five pooling methods (SpecialFirst, SpecialMean, InputLast, AllMean, DAAP; details in Appendix A.2) on instruction-following embedding and generic sentence embedding tasks. As shown in Table 5, DAAP achieves the best overall performance. Our analysis reveals several important findings that highlight the effectiveness of different pooling strategies.

First, methods relying solely on special tokens (SpecialMean, SpecialFirst) perform worst on instruction understanding, indicating their limited discriminative power for capturing complex instructional semantics. Second, InputLast shows strong instruction understanding but trails DAAP in generic tasks, suggesting that input semantics alone are insufficient for optimal performance across all evaluation dimensions. Third, the undifferentiated fusion approach of AllMean, which treats all tokens equally, leads to degraded performance compared to DAAP, demonstrating the importance of strategic token selection and weighted integration in embedding extraction.

Pooling Method	Inst.	Gen.
SpecM	55.22(−11.86)	59.10(−3.59)
SpecF	62.69(−4.39)	61.76(−0.93)
InputL	66.72(−0.36)	61.80(−0.89)
InputL+SpecF	66.51(−0.56)	61.51(−1.14)
AllMean	63.48(−3.60)	61.63(−1.03)
DAAP	67.08	62.69

Table 5. Performance comparison of different pooling strategies. SpecM: SpecialMean, InputL: InputLast, SpecF: SpecialFirst.

5.5. Instruction-Robustness Evaluation

As shown in Table 6, following (Peng et al., 2024), we measure the embedding model’s robustness to instruction perturbations. This experiment constructs three instruction sets for each clustering task on FewNerd via GPT-4: 10 *correct* (paraphrased from original), 10 *implicit* (indirectly

expressed), and 10 *incorrect* (task-irrelevant) instructions.

Let Δ_{ci} denote the average-score gap between *correct* and *incorrect* instructions, and Δ_{ii} the gap between *implicit* and *incorrect* ones. Larger deltas indicate stronger robustness.

Model	Δ_{ci}	Δ_{ii}
Instructor-Large	0.02	0.01
Llama2-7B-Chat	0.19	0.17
Inbedder	0.21	0.18
InstEmb	0.26	0.26

Table 6. Instruction-robustness on FewNerd tasks. We employ InstEmb based on Llama2-7b-chat.

InstEmb achieves the largest margins and $\Delta_{ci} \approx \Delta_{ii}$, confirming that it retains task understanding even under implicit wording.

5.6. Robustness Across Diverse Training Datasets

	Inst.	Gen.	Score	p-MRR
InstEmb _{absQA}	67.08	62.69	28.5	+15.6
InstEmb _{extQA}	66.51	62.39	27.3	+14.8
InstEmb _{MS-MARCO}	66.56	62.92	29.4	+15.7

Table 7. Performance across different training datasets, demonstrating consistent robustness.

To assess model robustness across different training datasets, we conduct experiments using two alternative datasets, both trained with the MSE objective.

The first dataset is an **Extractive QA Dataset (extQA)** containing approximately 150k examples from SQuAD_v2 (Rajpurkar et al., 2018) and NewsQA (Trischler et al., 2017), where answers are directly extracted from given contexts.

The second dataset is **MS-MARCO** (Nguyen et al., 2016), comprising around 80k examples characterized by longer answer sequences and more diverse, free-form generation patterns compared to extractive QA.

Table 7 presents the performance results. We observe minimal performance variation across different training datasets, indicating that our method maintains robust performance regardless of the underlying data paradigm. This stability enables straightforward utilization of diverse open-source datasets without requiring dataset-specific normalization or complex preprocessing.

6. Conclusion

We introduce **InstEmb**, a novel instruction following embedding framework that effectively captures both Input-Intrinsic and Output-Aware semantic information through representation self-distillation and supervised contrastive

learning. By leveraging learnable look-ahead tokens, InstEmb encodes rich instruction-conditioned semantics directly within a single prefilling step, significantly boosting embedding performance without additional decode latency. Experiments demonstrate that InstEmb achieves state-of-the-art results across multiple instruction following benchmarks. Our work provides an efficient yet powerful solution for instruction following embedding generation, paving the way for future research towards more effective and computationally efficient embedding methods.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Appendix

A.1. Implementation Details

We employ LLaMA-3-8B-Instruct as the backbone model. During knowledge distillation, we freeze the first 24 layers of the student model and all teacher model layers. InstEmb is trained for 1 epoch using Adam optimizer with learning rate 5×10^{-6} . The sequence length for look-ahead tokens is set to 8, which depends on data distribution. For ablation studies on training data, we set the look-ahead token length to 32 for MS-MARCO due to its longer average output length. However, during inference, we uniformly use a look-ahead length of 8 for all datasets to ensure consistency and fair comparison. For contrastive learning, we use dropout ratio 0.2 for view2 and temperature $\tau = 0.1$. Training uses maximum input sequence length 512, batch size 8, gradient accumulation steps 6, and bf16 precision on 8 NVIDIA H800 GPUs.

A.2. Pooling Method Details

- **SpecialFirst:** Selects the hidden state of the first special token as the representation:

$$e = h_{|x|+1}^S \quad (7)$$

- **SpecialMean:** Computes the mean embedding over all special tokens:

$$e = \frac{1}{L} \sum_{j=1}^L h_{|x|+j}^S \quad (8)$$

- **InputLast:** Utilizes the hidden state of the last input token:

$$e = h_{|x|}^S \quad (9)$$

- **AllMean:** Computes the mean embedding over the last input token and all special tokens:

$$e = \frac{1}{L+1} \left(h_{|x|}^S + \sum_{j=1}^L h_{|x|+j}^S \right) \quad (10)$$

- **DAAP:** Our proposed method explicitly integrates input and output semantics by averaging the last input token embedding (capturing input semantics) and the mean embedding of special tokens (capturing output semantics):

$$e = \frac{1}{2} \left(h_{|x|}^S + \frac{1}{L} \sum_{j=1}^L h_{|x|+j}^S \right) \quad (11)$$

A.3. Attention Pattern Visualization

As shown in Figure 3, we further conducted additional attention pattern visualizations and observed notable differences. Specifically, Llama3-8b-instruct exhibits a prominent *attention sink* (Xiao et al., 2024a) effect, where a substantial amount of attention fails to focus on meaningful information and instead defaults to the first position.

In contrast, after training, our InstEmb demonstrates a more sophisticated attention allocation strategy. Rather than concentrating attention indiscriminately on the first token due to uncertainty about relevant information, InstEmb selectively focuses on several semantically critical positions. These key positions include the end of the system prompt, which serves as a global instruction, and the end of the instruction. This selective attention pattern indicates that InstEmb has developed enhanced sensitivity to fine-grained instruction semantics, enabling more effective modeling of task-specific representations through dynamic association with instruction-relevant content.

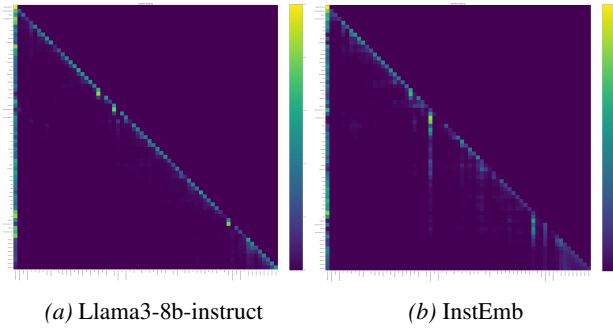


Figure 3. Comparison of attention patterns.

A.4. Semantic Alignment Visualization of Look-ahead Tokens

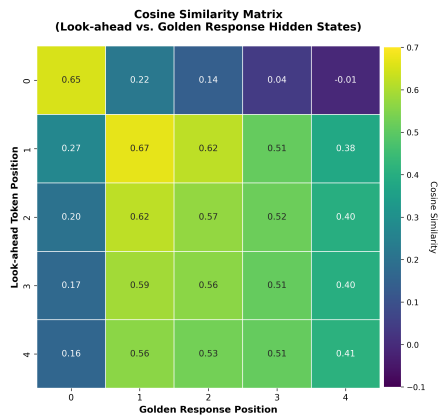


Figure 4. Cosine similarity matrices between the hidden states of the look-ahead token sequence and those of the golden output sequence.

To understand the semantic role of look-ahead tokens, we compute cosine similarities between the hidden states of: (1) the last input token and subsequent look-ahead tokens of InstEmb, and (2) the last input token and subsequent golden output tokens of LLaMA3.

As shown in Figure 4, we observe two key patterns: (1) the last input token embedding shows low similarity with other positions, indicating its focus on capturing Input-Intrinsic input semantics; (2) look-ahead tokens exhibit consistently higher similarity with the golden output sequence, demonstrating their role in capturing Output-Aware Semantics aligned with the intended output.